

Austrian MO Regionals 2025 — P1/4

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Let $n \geq 3$ be a positive integer. Furthermore, let $x_1, x_2, \dots, x_n \in [0, 2]$ be real numbers subject to $x_1 + x_2 + \dots + x_n = 5$. Prove the inequality

$$x_1^2 + x_2^2 + \dots + x_n^2 \leq 9.$$

When does equality hold?

Solution

Without loss of generality we can order the variables like so $x_1 \leq x_2 \leq \dots \leq x_n$. Denote $S = x_1^2 + x_2^2 + \dots + x_n^2$.

Claim 1: Let $i < n$ be the largest integer such that $x_i > 0$ and $x_{i+1} < 2$. Let r_{n-2}, r_{n-1}, r_n be non-negative reals with sum x_i . Denote $x'_k = x_k + r_k$, ($k = n-2, n-1, n$). Let the r_k be such that x'_k are in ascending order and $x'_k \in [0, 2]$. We claim that the transformation $x_i \rightarrow 0$ and $x_k \rightarrow x'_k$ never decreases the quantity S .

Proof: Let S be the original value of the sum of squares of (x_n) and let S' be the value after the aforementioned transformation. We note that

$$\begin{aligned} S' - S &= [(x_{n-2} + r_{n-2})^2 + (x_{n-1} + r_{n-1})^2 + (x_n + r_n)^2] - [(r_{n-2} + r_{n-1} + r_n)^2 + x_{n-2}^2 + x_{n-1}^2 + x_n^2] \\ &= 2(x_{n-2}r_{n-2} + x_{n-1}r_{n-1} + x_nr_n) - 2(r_{n-2}r_{n-1} + r_{n-2}r_n + r_{n-1}r_n) \\ &= 2(x_{n-2} - r_{n-1})r_{n-2} + 2(x_{n-1} - r_n)r_{n-1} + 2(x_n - r_{n-2})r_n \end{aligned}$$

and since $r_{n-2}, r_{n-1}, r_n \leq x_i \leq x_{n-2}, x_{n-1}, x_n$ the quantity $S' - S$ is ≥ 0 . $\square$

Now we can repeatedly make that transformation as given in claim 1 because it won't decrease S , in fact it can only increase S . After applying the transformation until we can't anymore we end up with $(x_1, x_2, \dots, x_{n-3}, x_{n-2}, x_{n-1}, x_n) = (0, 0, \dots, 0, 1, 2, 2)$. Indeed now we observe that

$$S \leq 1 + 4 + 4 = 9$$

as desired, with equality when $x_{n-1} = x_n = 2$ and $x_{n-2} = 1$. $\blacksquare$